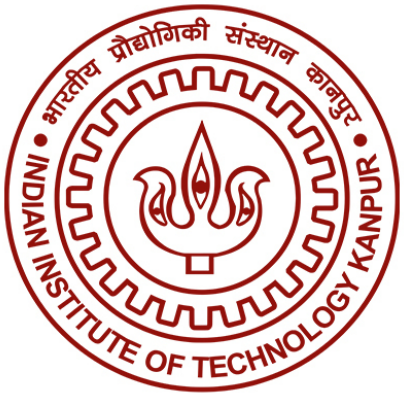



**SPA617: Radio Astronomy Lab**

# RF transmission, reflection, standing waves

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**Submitted by**

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## 1. Aim

To understand signal propagation, reflection and standing wave formation in a coaxial transmission line at radio frequencies and to determine important cable parameters experimentally.

## 2. Objectives

The objectives of the experiment are:

1. To determine the phase velocity and velocity factor of a signal in a coaxial cable, by measuring the frequency spacing between the consecutive minima produced the reflection caused in an open ended cable.
2. To study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load.
3. To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched with to a 50 Ω terminnation, indicating that its characteristic impedance is close to 50 Ω.
4. To measure the insertion loss and attenuation per unit length of the coaxial cable.
5. To calculate the inductance and capacitor per unit length of the coaxial cable using the measured phase velocity and the approximate characteristic impedance of 50 Ω

## 3. Theory

A coaxial cable is a transmission line used to guide electromagnetic signals from one point to another. At radio frequencies, the signal propagates mainly as a TEM wave, and its velocity depends on the dielectric material between the conductors. When a travelling wave reaches a load whose impedance is different from the characteristic impedance of the cable, part of the signal is reflected. The incident and reflected waves interfere with each other, producing maxima and minima in the measured signal amplitude. By varying the frequency, the phase of the reflected wave changes, causing a periodic variation in amplitude. For an open-ended cable, the frequency difference between two consecutive maxima or minima can be used to determine the phase velocity:

$$v_p = 2l\Delta f$$

where l is the cable length and $\Delta f$ is the frequency spacing. The velocity factor is given by

$$VF = \frac{v_p}{c}$$

The voltage reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

where $Z_L$ is the load impedance and $Z_0$ is the characteristic impedance of the cable. For an open circuit, $|\Gamma| = 1$, while for a matched load $Z_L = Z_0$ , $\Gamma = 0$. The inductance and capacitance per unit length of a low-loss transmission line are related to the characteristic impedance and phase velocity by

$$L' = \frac{Z_0}{v_p}$$

,

$$C' = \frac{1}{Z_0 v_p}$$

The cable also introduces attenuation as the signal propagates. The insertion loss is determined by comparing the measured power with the cable to the refer- ence power without the cable. The attenuation per unit length is then obtained by dividing the insertion loss by the cable length.

## 4. Equipment

The following equipment was used during the experiment

- Radio-frequency signal generator
- Spectrum analyzer
- RF Tee connector
- Oscilloscope / DSO
- Coaxial cables
- 50 Ω termination

## 5. Experimental Setup

The RF signal generator is connected to a Tee connector. One arm of the Tee connector is connected to the coaxial cable under test, while the other arm is connected to the

DSO. The end of the coaxial cable is terminated as required. The frequency of the signal generator is varied, and the signal amplitude on the DSO is observed. The frequencies corresponding to two consecutive minima are noted as f1 and f2 .The frequency difference is calculated as

$$\Delta f = f_2 - f_1$$

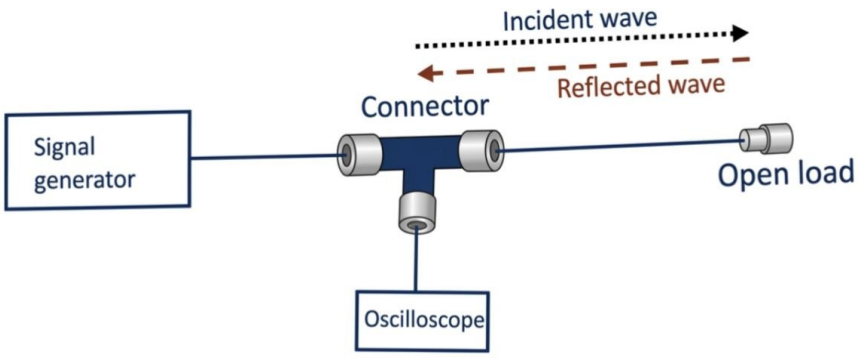



Figure 1: Schematic representation of the experimental setup.

As shown in Figure 2, the instruments are connected according to the experimental configuration. The relevant parameters are then varied systematically while recording the corresponding measurements.

## 6. Observation

Before starting the setup we carefully measured the coaxial cable which turned out to be 92.5 cm. Then after connecting all the cables with the Tee connector we observed the consecutive minimas using the Digital storage Oscilloscope (DSO). The frequency difference between those minimas was

$$\Delta f = f_2 - f_1 = (172 - 57) = 115MHz$$

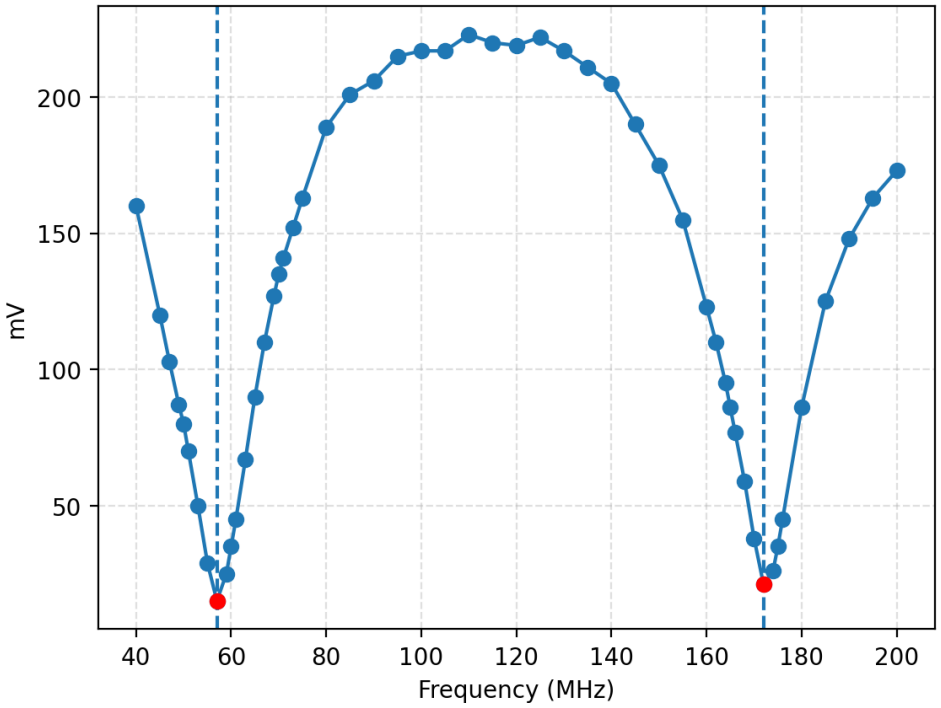



Figure 2: Consecutive minimas

After finding the frequency difference we can use that to find phase velocity

$$v_p = 2l\Delta f$$

$$v_p = 2 \times 92.5 \times 115 \ m/s$$

$$v_p = 2.127 \times 10^8 \ m/s$$

Now with calculated phase velocity we can calculate the velocity factor-

$$VF = \frac{v_p}{c}$$

$$VF = \frac{2.127 \times 10^8}{3 \times 10^8}$$

$$VF = 0.7091$$

In the next part of the experiment we checked the characteristic impedance. For that we first setup the Tee connector to an open circuit, then reconnected with a 50 Ω. Retaining the same frequency as before to carefully determine the $V_{max,open}$ and $V_{min,open}$ for calculating the reflection coefficient. The same thing we have done with the 50 Ω termination load to note the difference in reflection coefficient.

Reflection coefficient for open circuit,

$$|\Gamma| = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{1.7 - 0.048}{1.7 + 0.048} = 0.95.$$

Reflection coefficient for 50  load,

$$|\Gamma| = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} = \frac{1.12 - 0.93}{1.12 + 0.93} = 0.09.$$

Table 1: Reflection coefficient for open and 50  load

| **Termination** | $\mathbf{V}_{\max}$ (V) | $\mathbf{V}_{\min}$ (V) | $\|\mathbf{\Gamma}\|$ |
|---|---|---|---|
| Open circuit | 1.7 | 0.048 | 95% |
| 50 Ω load | 1.12 | 0.93 | 9% |

With this substantial amount of reduction in reflection coefficient we can conclude that the cable is approximately matched with to the 50 Ω termination load. The calculated characteristic impedance turned out to be

The characteristic impedance as calculated taking both $-\Gamma$ and $+\Gamma$,

$$Z_0 = Z_l \times \frac{1 - \Gamma}{1 + \Gamma}.$$

$$Z_0 = 50 \times \frac{1 - 0.09}{1 + 0.09} = 41.7\ \Omega$$

$$Z_0 = Z_l \times \frac{1 + \Gamma}{1 - \Gamma}.$$

$$Z_0 = 50 \times \frac{1 + 0.09}{1 - 0.09} = 59.8\ \Omega$$

Inductance per unit length,

$$\mathrm{L}' = \frac{\mathrm{Z_0}}{\mathrm{v_p}} = \frac{41.7}{2.13 \times 10^8}\ \mathrm{H/m} = 19.59 \times 10^{-8}\ \mathrm{H/m}.$$

Capacitance per unit length,

$$\mathrm{C}' = \frac{1}{\mathrm{Z_0} \times \mathrm{v_p}} = \frac{1}{41.7 \times 2.13 \times 10^8}\ \mathrm{F/m} = 0.011 \times 10^{-8}\ \mathrm{F/m}.$$

For the cable attenuation per unit length the spectrum analyser has been used to record

reference power $P_{ref}$ and $P_{cable}$. Calculated attenuation per unit length,

$$\alpha = \frac{\mathrm{P_{ref}} - \mathrm{P_{cable}}}{\mathrm{l}} = \frac{0 - (-0.65)}{0.925}\ \mathrm{dB/m} = 0.7\ \mathrm{dB/m}.$$

Table 2: Attenuation of the cable

| **Frequency(MHz)** | $P_{ref}(dBm)$ | $P_{cable}(dBm)$ | $\alpha\ (dB/m)$ |
|---|---|---|---|
| 60 | 0.00 | -0.65 | 0.7 |

Table 3: Experimental results for coaxial cable

| **Quantity** | **Symbol** | **Measured/Calculated Value** |
|---|---|---|
| Cable length (m) | $\ell$ | 92.5 cm |
| Frequency spacing (Hz) | $\Delta f$ | 115 MHz |
| Phase velocity (m/s) | $v_p$ | $2.127 \times 10^8$ m/s |
| Velocity factor | $v_p/c$ | 0.7091 |
| Approximate characteristic impedance ($\Omega$) | $Z_0$ | $41.7, 59.8$ |
| Inductance per unit length (H/m) | $L'$ | $19.59 \times 10^{-8}$ H/m |
| Capacitance per unit length (F/m) | $C'$ | $0.011 \times 10^{-8}$ F/m |
| Attenuation (dB/m) | $\alpha$ | 0.7 dB/m |

### 6.1 Bonus part

Table 4: For calculation of unknown length of cable

| **Minima** | **Frequency(MHz)** | **Voltage(V)** |
|---|---|---|
| 1st | 52 | 120 |
| 2nd | 87 | 170 |

Frequency difference $\Delta f = (87 - 52)$ MHz $= 35$ MHz.

Length of the unknown cable,

$$l = \frac{v_p}{2 \times \Delta f} = \frac{c \times \mathrm{VF}}{2 \times \Delta f} = \frac{3 \times 10^8 \times 0.71}{2 \times 35 \times 10^6}\ \mathrm{m} = 3.04\ \mathrm{m}.$$

**Error calculation for the unknown length**

$$\%\ \mathrm{Error} = |\frac{3.06 - 3.04}{3.04}| \times 100\%$$

$$\%Error = 0.65$$

## 7. Results and Discussion

- In this experiment the characteristics of a coaxial cable were studied using the RF signal generator, Tee connector and DSO. The coaxial cable be- haves as transmission line at high frequency and the signal doesn't travel simply from source to load. When load is not matched to the characteristic impedance of the cable, a part of the signal is reflected back towards the source. By varying the frequency and observing the consecutive minimas on the DSO the propagation velocity and other parameters of coaxial cable like velocity can be determined experimentally.

- We observed that changing the termination changes the reflected wave. In the case for short circuit the reflection coefficient is $-1$, which indicates the reflected wave is of same magnitude but with opposite phase. That is why it creates a standing wave producing successive minima and maxima. Same goes for the open ended circuit with only difference being that its maxima and minima position are interchanged with position of maxima minima for short circuit. In case of 50 $\Omega$ load the reflection coefficient substantially drops to low value causing almost no reflected wave for interference.

- As the velocity factor term stays constant for every cable, it can used to determine the length of a unknown cable by simply observing the frequency difference. In our case the given unknown cable's length turned out to be 305.7 cm which was well under the actual measured length (306.8 cm).

- The effect of termination load is also studied for calculating impedence and conductance per unit length.